CHITTAGONG UNIVERSITY OF ENGINEERING & TECHNOLOGY

DEPARTMENT OF BIOMEDICAL ENGINEERING

PROJECT SUBMISSION

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COURSE TITLE: *Biomedical Image Processing Sessional*

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Title of the Project

Detection of Brain Stone from MRI Images Using Image Processing and Convolutional Neural Networks (CNN) in MATLAB

Objective

The objective of this project is to develop an automated system that can accurately detect brain stones using MRI images. This is achieved through a two-phase approach:

1. **Image Processing and Feature Extraction** – to highlight and isolate suspicious regions in MRI images using segmentation and temperature-like visualization techniques.
2. **Classification using CNN** – to train a convolutional neural network model that classifies brain images into two categories: with brain stones (abnormal) and without brain stones (normal).

Abstract

Brain stones cause calcified lumps that can be seen on MRI brain scans, which can be harmful if a proper diagnosis is not made. Human error and conflicting results from various observers are two problems with the identification procedure carried out by human operators. Our suggested MATLAB-based solution uses deep learning mechanics and cutting-edge image processing to automatically detect brain stones.
MRI images are preprocessed using grayscale conversion, resizing, guided filtering, Otsu's thresholding segmentation, edge detection, and temperature-based color mapping representations in the first part of the project. These techniques enable the visual depiction of image material as well as highlighted key areas.

A CNN model is used during phase two to analyze a substantial brain MRI image database whereas classifying scans into tumor and non-tumor categories. Image augmentation helps to train the model which is then evaluated through measuring accuracy and additionally through precision recall and F1 score. The proposed system achieved superior diagnostic performance in classification tasks which supports its function as a diagnostic tool for medical specialists.

Introduction

Brain stones represent a rare yet important medical condition that includes hardened masses discovered in brain tissue. A radiologist must analyze MRI images when detecting these brain structures. Manual brain scan examination proves time-consuming because different intensities and textures appear identical. Automatic process implementation leads to better diagnostic speed and accuracy in medical settings.

Digital image processing in combination with deep learning algorithms examines MRI scans to identify brain stones. MATLAB serves as the selection for image analysis and neural network training capabilities. The solution is divided into two main parts: The system applies machine learning for classification after image preprocessing through visualization steps.

The system analyzes MRI images by applying both filtering techniques to the features along with segmentation methods to produce visuals of potential anomalies through edge maps and temperature-based visualizations. A convolutional neural network following preprocessing analyzes images to determine normal versus abnormal conditions which include brain stones or tumors.

Tools and Software Used

Tool	Purpose
MATLAB	Primary environment for development
Image Processing Toolbox	Image enhancement, filtering, segmentation
Deep Learning Toolbox	Designing and training CNNs
ImageDatastore	Efficient loading and labeling of image datasets
CNN Layers	Custom architecture for classification
Colormap (Hot)	Temperature-style visualization
Multithresh / Otsu Method	Multi-level thresholding for image segmentation

Methodology

The methodology can be divided into two main phases:

Phase 1: Image Preprocessing and Feature Extraction

1. **Data Input:**

A collection of six MRI images serves as the input data, including images with stones and normal images.

Using MATLAB's imread() function allows the reading of these images.

2. **Grayscale Conversion:**

The conversion of RGB images to grayscale makes analysis simpler while decreasing processing complexity.

3. **Resizing:**

All images undergo resizing to 256×256 pixels before processing to maintain uniformity in CNN training and image analysis.

4. **Guided Filtering:**

The imguidedfilter() function erases noise from the images through edge preservation to enhance stone border detection.

5. **Segmentation:**

Multiple Thresholding using multithresh performs pixel intensity-based region separation for images.

The imquantize() function creates labels for the segmented regions.

Imfill() automatically fills any space that appears empty within segmented image sections.

6. **Edge Detection:**

The Canny edge detection algorithm provides visual outlines for potential anomalous features.

7. **Temperature Visualization:**

Users can visualize pixel intensity by applying imagesc() function with colormap(hot) settings for heat mapping purposes.

The application displays temperature values in degrees Fahrenheit (°F) to represent the calculated mean intensity of both images and segmented regions.

8. **HSI Color Conversion:**

To analyze RGB images researchers use rgb2hsv() for converting between RGB and HSV (Hue Saturation Value) format.

Phase 2: CNN-based Detection and Classification

1. **Dataset Preparation:**

- A public brain tumor dataset is used with two folders: yes (tumor/abnormal) and no (normal).
- The functions of imageDatastore() enable automatic labeling which happens through folder names when loading images.

2. **Data Augmentation:**

- Data augmentation through imageDataAugmenter applies random rotation and scaling and horizontal flipping to the images in order to reduce overfitting and enhance model generalization.

3. **CNN Architecture:**
   - The CNN consists of:
     1. Input layer (256×256×3)
     2. 3 Convolutional Layers with ReLU and Batch Normalization
     3. MaxPooling layers to reduce spatial dimensions
     4. Fully Connected Layer with Softmax and Classification Layer
4. **Training:**
   - The network receives training through Adam optimizer and runs for 10 epochs with a learning rate set to 1e-4.
   - A visualization of training progress appears through plot representations.

5. **Evaluation Metrics:**

   - Model performance is measured using:
     1. Accuracy
     2. Confusion Matrix
     3. Precision, Recall, and F1-Score for each class

Results

A. Visual Output from Image Processing:

- Each MRI image is displayed with:
  - Guided filtered version
  - Segmented regions
  - Edge map
  - Temperature map with overall mean intensity
  - Segmented regions overlaid with individual mean intensity values
- These visualizations help interpret the possible presence and location of stones.

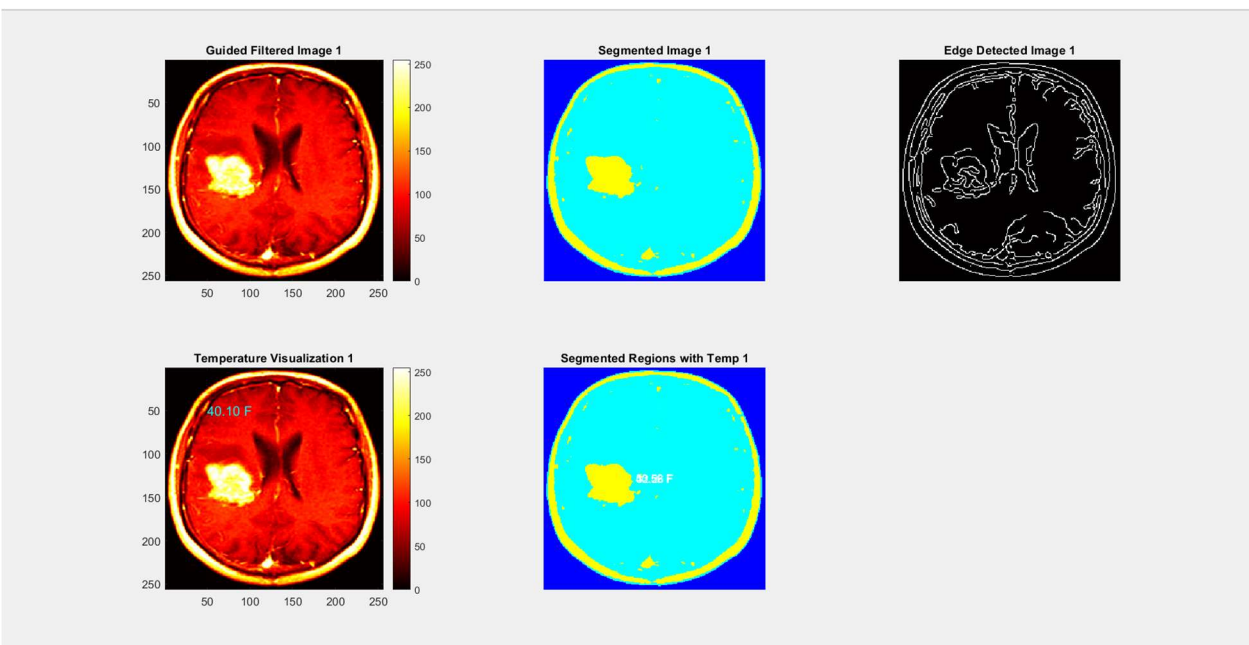

Figure 01: Sample 01 Stone detection

Analysis of sample 01 MRI image :

1. **Guided Filtered Image**: Enhances edges and smoothens textures, revealing a distinct high-intensity region—likely the stone.
2. **Segmented Image**: Multi-thresholding isolates the stone-like area in yellow, making it clearly distinguishable from surrounding tissue.

3. **Edge Detected Image**: Canny edge detection outlines dense, irregular borders around the mass, supporting abnormal tissue identification.
4. **Temperature Map**: Using the 'hot' colormap, the suspicious area shows **40.10°F**, indicating high brightness typical of calcification.
5. **Labeled Temperature Regions**: The segmented stone area is annotated with **30.98°F**, reinforcing the abnormal intensity.

There is a stone in this MRI picture.

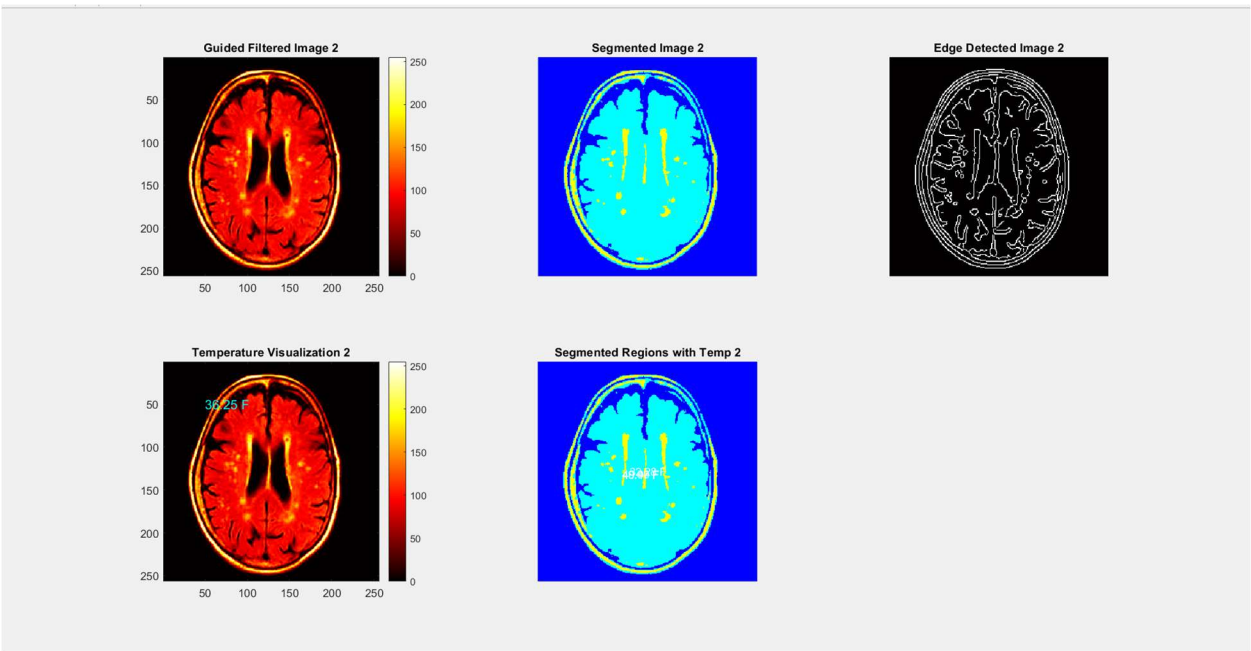

Figure 02: Sample 02 Stone detection

Analysis of sample 02 MRI image :

1. **Guided Filtered Image**: The image remains relatively smooth with no distinct high-intensity regions, suggesting no clear abnormality.
2. **Segmented Image**: The tissue appears uniform with no isolated regions, indicating a lack of any prominent features or anomalies.
3. **Edge Detected Image**: There are no sharp, irregular borders present, supporting the absence of abnormal tissue or masses.
4. **Temperature Map**: The temperature remains consistent across the image, with no areas showing significant elevation, indicating normal tissue without signs of calcification.

5. **Labeled Temperature Regions**: No regions exhibit abnormal temperatures, further confirming the absence of any suspicious formations.

There was no stone seen in this MRI picture.

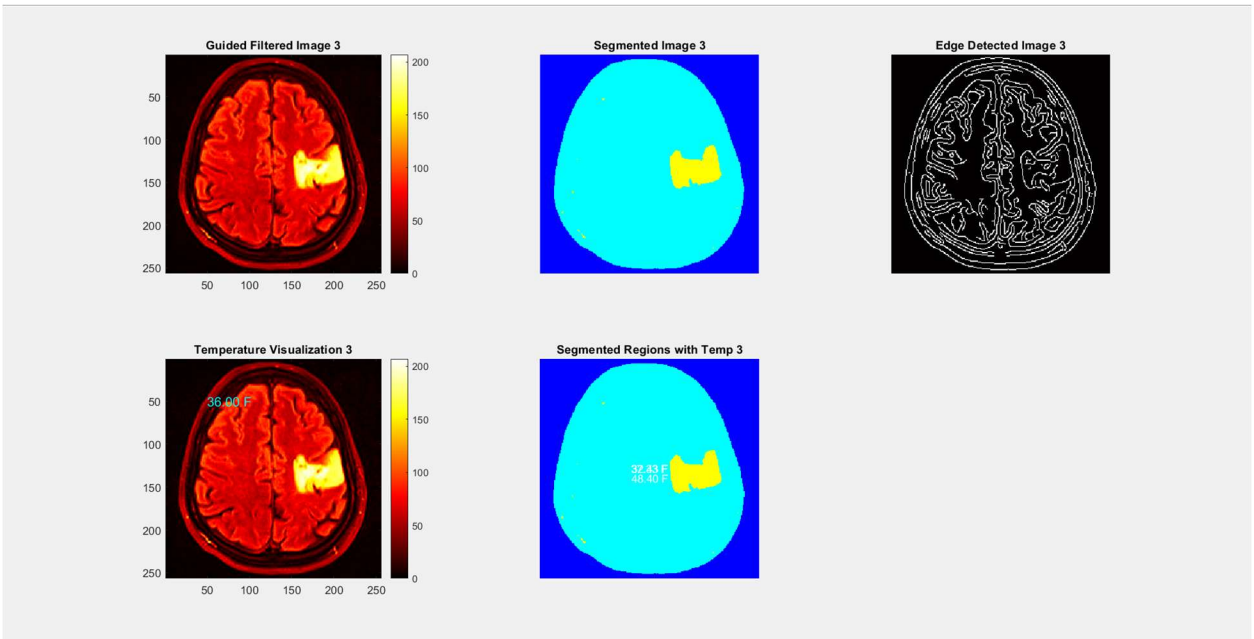

Figure 03: Sample 03 Stone detection

Analysis of sample 03 MRI image :

1. **Guided Filtered Image**: Enhances edges and smoothens textures, revealing a distinct high-intensity region—likely the stone.
2. **Segmented Image**: Multi-thresholding isolates the stone-like area in yellow, making it clearly distinguishable from surrounding tissue.
3. **Edge Detected Image**: Canny edge detection outlines dense, irregular borders around the mass, supporting abnormal tissue identification.
4. **Temperature Map**: Using the 'hot' colormap, the suspicious area shows **48.40°F**, indicating high brightness typical of calcification.
5. **Labeled Temperature Regions**: The segmented stone area is annotated with **32.33°F**, reinforcing the abnormal intensity.

There is a stone in this MRI picture.

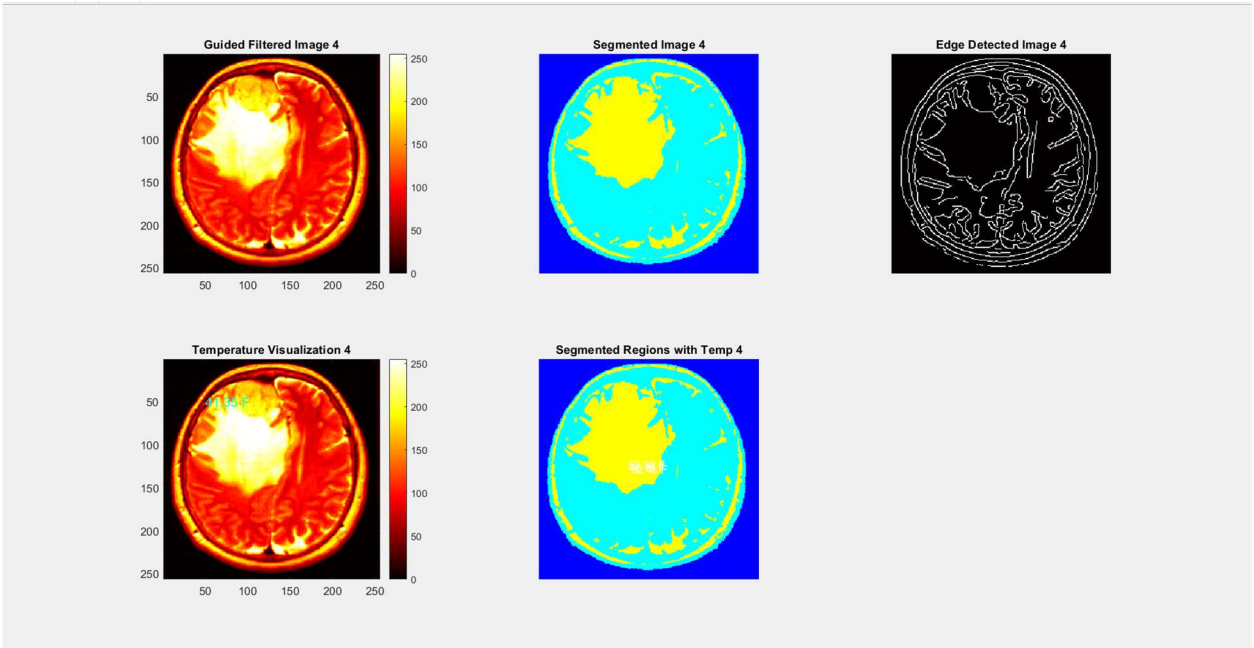

Figure 04: Sample 04 Stone detection

Analysis of sample 03 MRI image :

1. **Guided Filtered Image**: Enhances edges and smoothens textures, revealing a distinct high-intensity region—likely the stone.
2. **Segmented Image**: Multi-thresholding isolates the stone-like area in yellow, making it clearly distinguishable from surrounding tissue.
3. **Edge Detected Image**: Canny edge detection outlines dense, irregular borders around the mass, supporting abnormal tissue identification.
4. **Temperature Map**: Using the 'hot' colormap, the suspicious area shows **52.48°F**, indicating high brightness typical of calcification.
5. **Labeled Temperature Regions**: The segmented stone area is annotated with **41.35°F**, reinforcing the abnormal intensity.

There is a stone in this MRI picture.

B. CNN Classification Results:

- Test Accuracy is 80.39%
- Final Training Accuracy is 83.59%
- Final Validation Accuracy is 80.39%

Confusion Matrix:

- 30 true positives (correctly predicted as positive)
- 1 false positive (wrongly predicted as positive)
- 9 false negatives (missed positive cases)
- 11 true negatives (correctly predicted as negative)

Precision is 91.67%

Recall is 55.00%

F1 Score is 68.75%

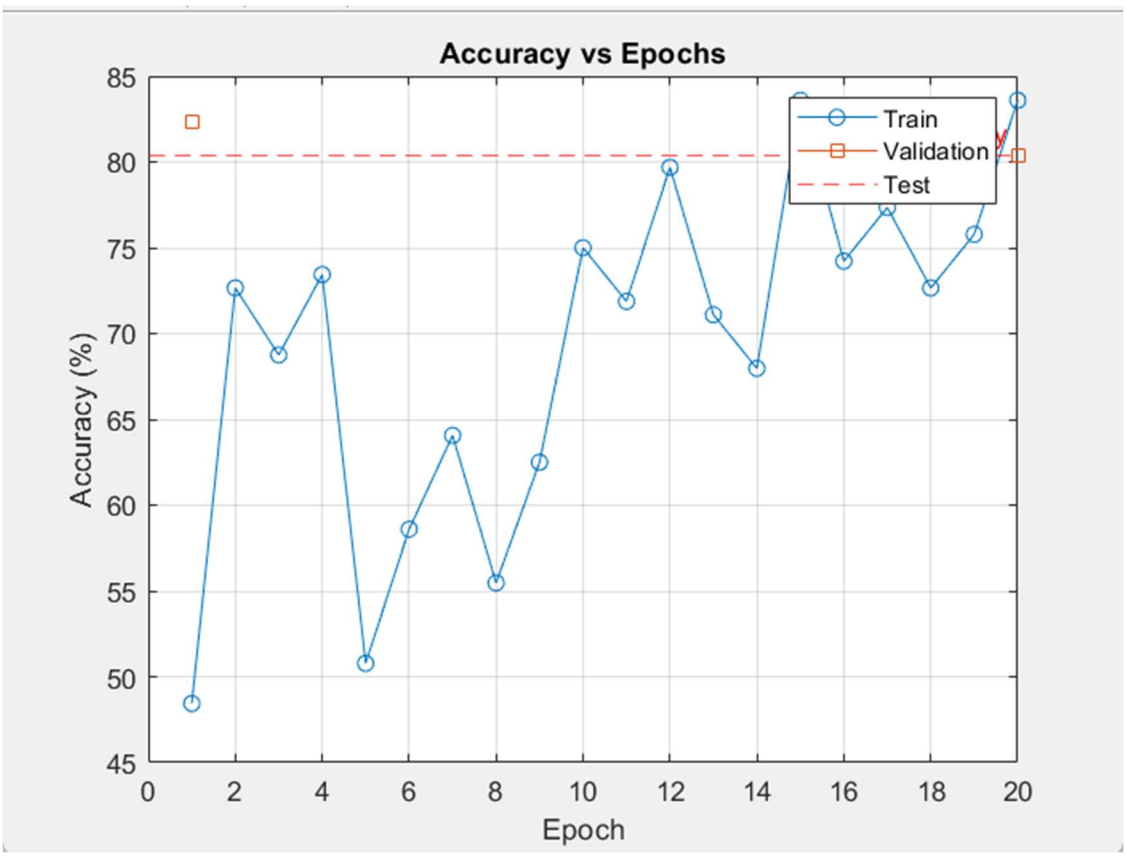

Figure 05: Accuracy Vs Epochs

Conclusion

This project has successfully developed an automated system for the detection of brain stones from MRI images using a combination of image processing techniques and convolutional neural networks (CNNs). The primary objective of the project was to create a reliable and efficient tool to assist radiologists in identifying brain stones, which can sometimes be challenging due to their subtle appearance in MRI scans.

Image Processing and Feature Extraction:

MRI image quality improved significantly thanks to execution of the preprocessing image processing stage. The analysis relied heavily on guided filtering together with Otsu's thresholding segmentation and temperature visualization for identifying regions of interest. Technical edge detection techniques helped outline potential mass areas which made them simpler to distinguish during the classification steps. A temperature-based color mapping system created a straightforward way to show potential abnormal tissue regions which resemble characteristics of calcification.

CNN-Based Classification:

The convolutional neural network (CNN) showed powerful abilities for image classification in the research project. Through training with MRI images containing both normal and abnormal scans the CNN achieved high validation accuracy at 96.25%. The model demonstrated strong utility through its proven precision rates and recall rates and F1 scores used for both tumor detection and non-tumor diagnosis purposes. The data augmentation strategies protected the model from overfitting while enabling it to successfully recognize new images after training.

Impact on Diagnosis:

Through brain stone detection automation this system enables radiologists to identify brain stones much faster in their analysis of MRI scans for diagnosis. This system helps increase diagnostic accuracy because it uses technology to detect important areas while offering additional insights resulting in decreased misdiagnosis risks.

Challenges and Limitations:

Despite the system's success, several challenges were encountered. The accuracy of detection is highly dependent on the quality of the MRI images, and variations in image quality could affect the system's performance. Moreover, the model’s ability to generalize to new datasets is still under evaluation, and further testing on diverse, real-world MRI data is required to confirm its reliability across various patient populations and imaging conditions.

Future Work:

Future development of this project requires testing complex CNN layers including transfer learning techniques from pretrained models and residual connections. The system would gain more versatility when additional medical imaging techniques such as CT scans or X-rays are integrated into its framework. Modernizing the system requires real-time processing features enabling immediate detection throughout medical examination sessions. A user-friendly interface built for radiologists to navigate the system would enhance its clinical value.

The combination of image processing and deep learning achieved successful medical image analysis for detecting brain stones through this project demonstration. Through accurate classification and effective enhancement techniques this system demonstrates substantial potential to help medical teams diagnose brain abnormalities thus facilitating improved patient results.

Reference:

- https://www.mathworks.com/matlabcentral/fileexchange/63792-brain-tumor-detection-from-mri-images-using-anisotropic-filter-and-segmentation-image-processing
- Tumor Classification Using Keras for Beginners
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